SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL

CLASS: XII **SUB: COMPUTER SCIENCE**

LAB QUESTIONS-21 TO 25

PROGRAM:21

Write a menu-driven Python program using mysql.connector to perform basic initialization and inspection tasks.

Create a database STUDENTDB and implement functions for:

• Create Table: Create the STUDENT_RECORDS table with the 6 columns listed above.

• Insert Records: Add the 5 predefined records into the table.

• Display All: Fetch and display the records

PROGRAM:22

write a menu-driven program that handles record modification and basic statistics. Create user-defined functions for:

• Update Marks: Prompt the user to enter a student ID, a new TOTAL, and a new GRADE to modify that specific student's row.

• Count Records: display the total number of students currently present in the table

PROGRAM:23

Create a menu-driven program that isolates targeted sub-sections of your students. Write functions for:

• Filter Top Scorers: Search and display records of all students who scored a TOTAL greater than or equal to 400 marks.

• Update Section: Transfer a student to a different group by updating their SECTION attribute based on their unique ID.

• Delete Low Performers: Automatically drop any student record from the table if their GRADE matches a user-specified lower tier (like 'D').

PROGRAM:24

Write a menu-driven program to filter and clean data within the existing STUDENT_RECORDS table. Build functions for:

• Delete Student: Remove a student's row based on a user-specified ID.

• Filter by Class: Prompt the user to type a class name (e.g., '12') and display only the students who belong to that specific class

PROGRAM:25

Create a brand new table named FACULTY_RECORDS inside the same database with the fields: EMP_ID (Primary Key), NAME, DEPARTMENT, CLASS_TEACHER_OF, SALARY, and STATUS. Add 5 baseline teacher rows, then write code for a menu containing:

• Filter Department: Input a department name (e.g., 'Science') and list all teachers working under it.

• Update Salary: Modify a teacher's SALARY using their unique EMP_ID.

• Delete Inactive: Remove faculty records where the STATUS is set to 'Left' or 'Retired'.